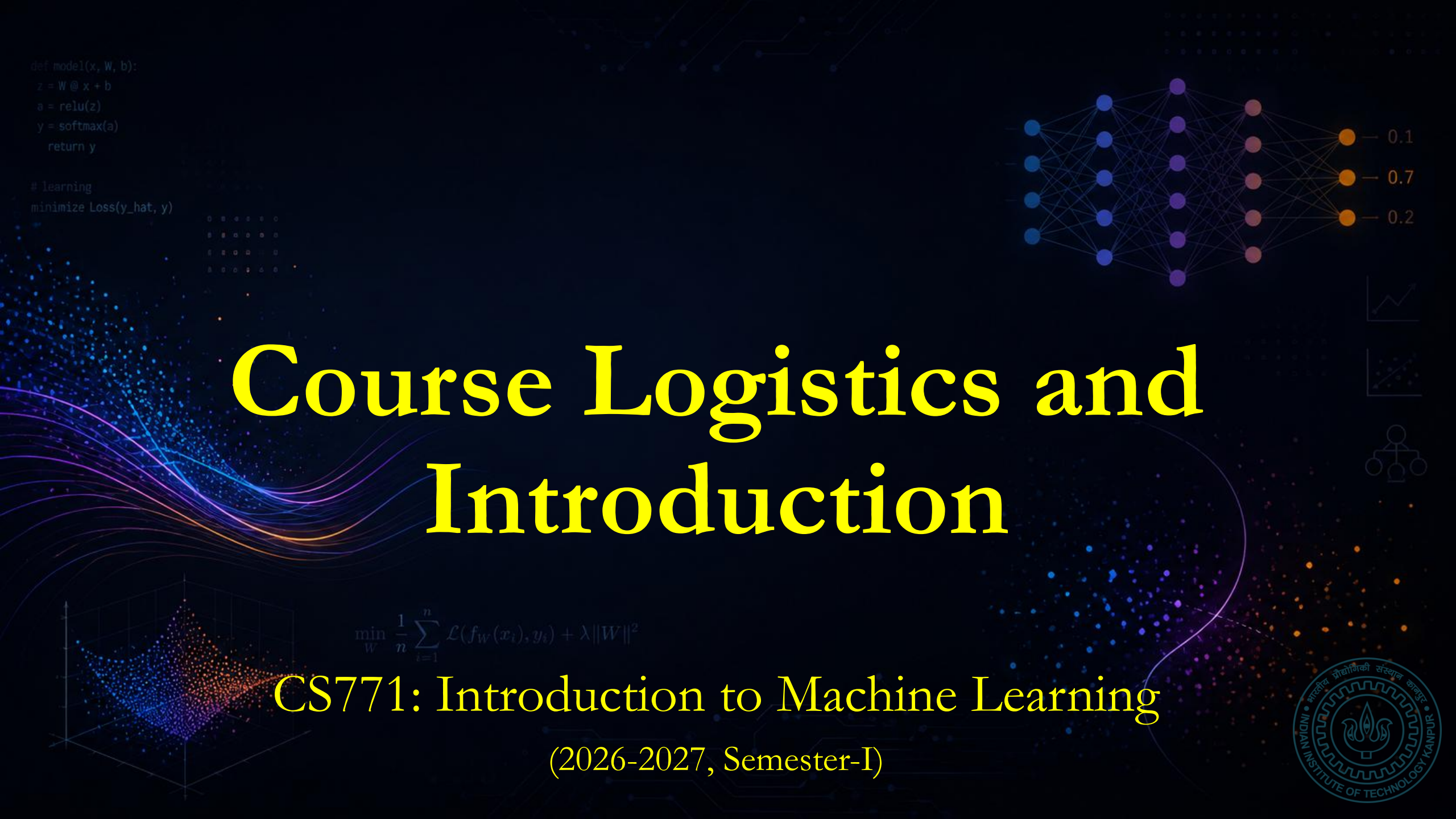


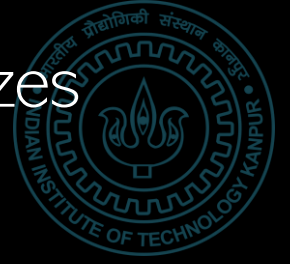
Course Logistics and Introduction

CS771: Introduction to Machine Learning

(2026-2027, Semester-I)

Course Logistics

- Timing and Venue: Wed/Fri 6:15-7:30pm, L-20
- Course webpage: https://tinyurl.com/cs771-a26 (slides, readings, etc)
- Online discussion/QA: **Piazza** (https://tinyurl.com/cs771-a26-piazza)
- Instructor's contact email: piyush@cse.iitk.ac.in, office: RM-502 (CSE dept)
  - Prefix email subject with CS771, else might get ignored
  - Piazza is encouraged for course-related matters (also has private messaging and DMs)
  - Office hours: By appointment (can meet in person or online)
- TAs: Their contact details and office hours details shared soon
- Unofficial auditors are welcome. However, can't participate in exams/quizzes
  - Can attempt homeworks, quizzes, exams on their own. Won't be graded



Workload and Grading Policy

Quizzes will be closed-book. For exams, one A4 size cheat-sheet will be allowed



- 12 credits course
  - There is a lab component (in form of take-home weekly programming practice/assignment)
- 4 quizzes: 30%
- 2 programming assignments/mini-projects: 20%
  - Writeups must be prepared in PDF using the provided LaTeX template
  - Knowledge of Python programming is assumed
  - To be done in groups of 5 students. Form your groups NOW
- Mid-sem exam: 20%
- End-sem exam: 30%
- Quiz dates: Tentative dates will be announced by next week
  - Quiz timing and venue: will be announced closed to the quiz date
- Mid-sem and end-sem exam dates: As per DOAA announcements

Preparation Requirements, Reference Books

- Brush up maths (linear algebra, probability, and statistics) concepts if rusty
  - May find these refresher slides helpful: https://web.cse.iitk.ac.in/users/piyush/courses/ml-autumn23/slides/maths_refresher_slides.pdf
  - Relevant background will also be introduced as and when required

- Learn some basic Python if not familiar already
  - May find this Colab tutorial helpful (brief but focused on ML): https://colab.research.google.com/drive/1qmN_40NUdp6V5CQik_SRfcQ9H9IB1oZr
  - This link provides other general resources for learning Python programming: www.geeksforgeeks.org/python-programming-language/

- Learn basic LaTeX for preparing PDFs
  - Useful links available here: www.overleaf.com/learn/latex/Tutorials

- No reference book for this course. We will provide you the reference readings on various topics whenever required



The AI Revolution..

Even this intro lecture's slides are created by taking some help from AI – basically by the instructor prompting an LLM to generate some of the figures used here 😊



AI is Everywhere

From everyday apps to scientific breakthroughs, AI is transforming the world around us.

Everyday Life
Chat, search, translate, recommendations, voice assistants, and more.

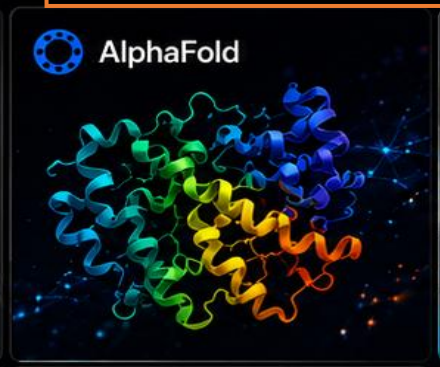
Science & Medicine
Solving protein structures, discovering drugs, diagnosing diseases.

Transportation & Robotics
Self-driving cars, smart robots, drones, and beyond.

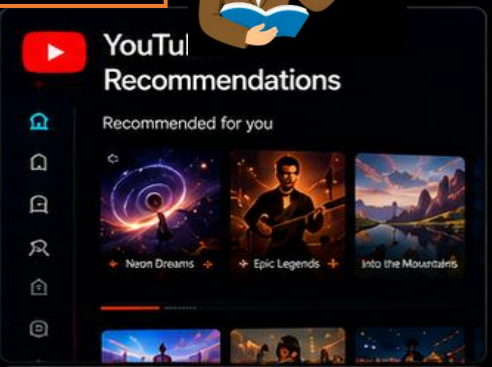
Creativity & Media
Generating images, videos, music, and designing the future.

Software & Business
Coding assistants, fraud detection, customer support, and automation.

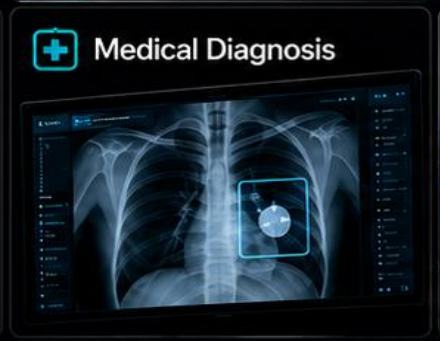
































The AI Revolution: Three Key Enablers

1 MASSIVE DATASETS

More data covers more examples, variety and patterns from the real world.



Web scale data | Trillions of examples | Continuously growing

More and better data leads to better learning.

2 POWERFUL HARDWARE (GPU)

Massive parallelism enables training of large models in feasible time.

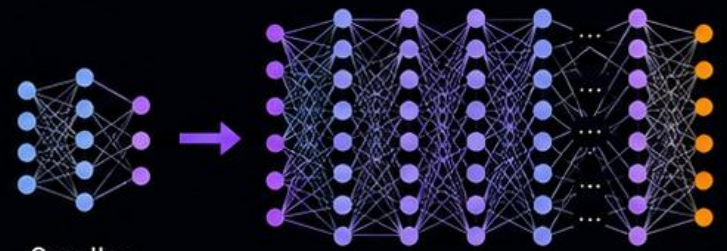


Thousands of cores | High throughput & memory | Distributed at scale

More compute power enables larger and faster training.

3 BETTER LEARNING ALGORITHMS

Advances in algorithms, architectures and optimization drive better performance.



Better architectures | Improved optimization | New ideas & regularization

Smarter algorithms learn more from the data.

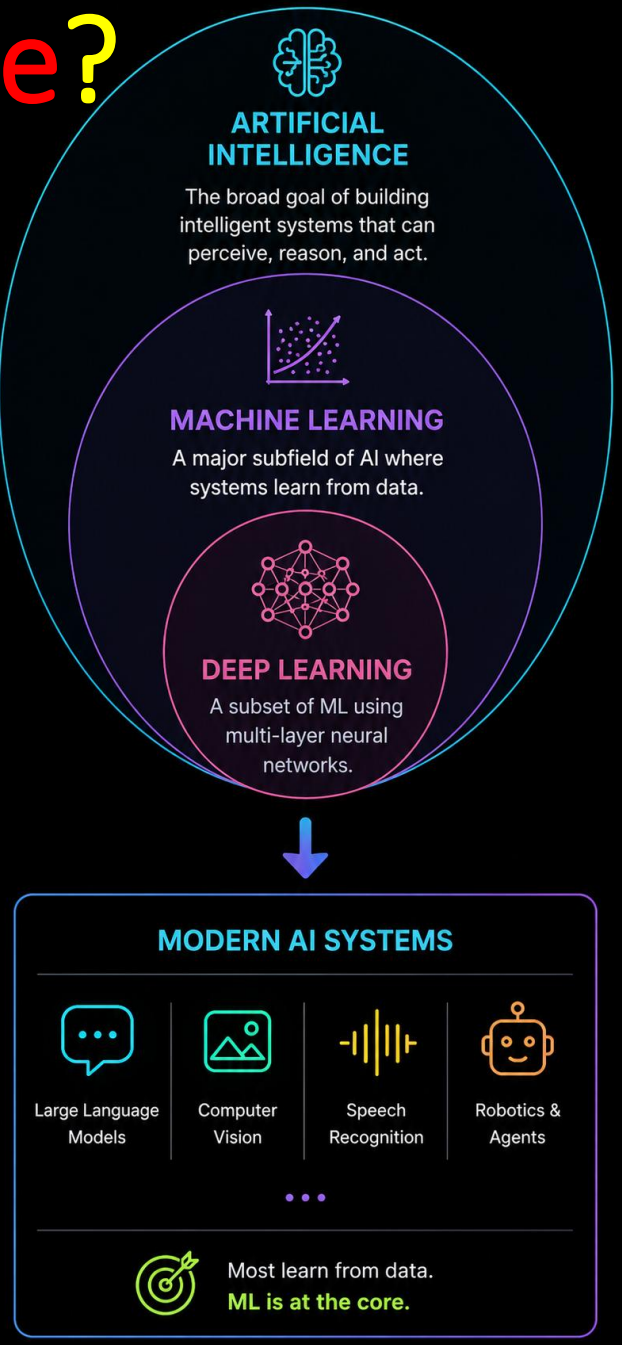
Big data + Powerful compute + Better algorithms = Modern AI revolution

Individually important. Together, transformational.

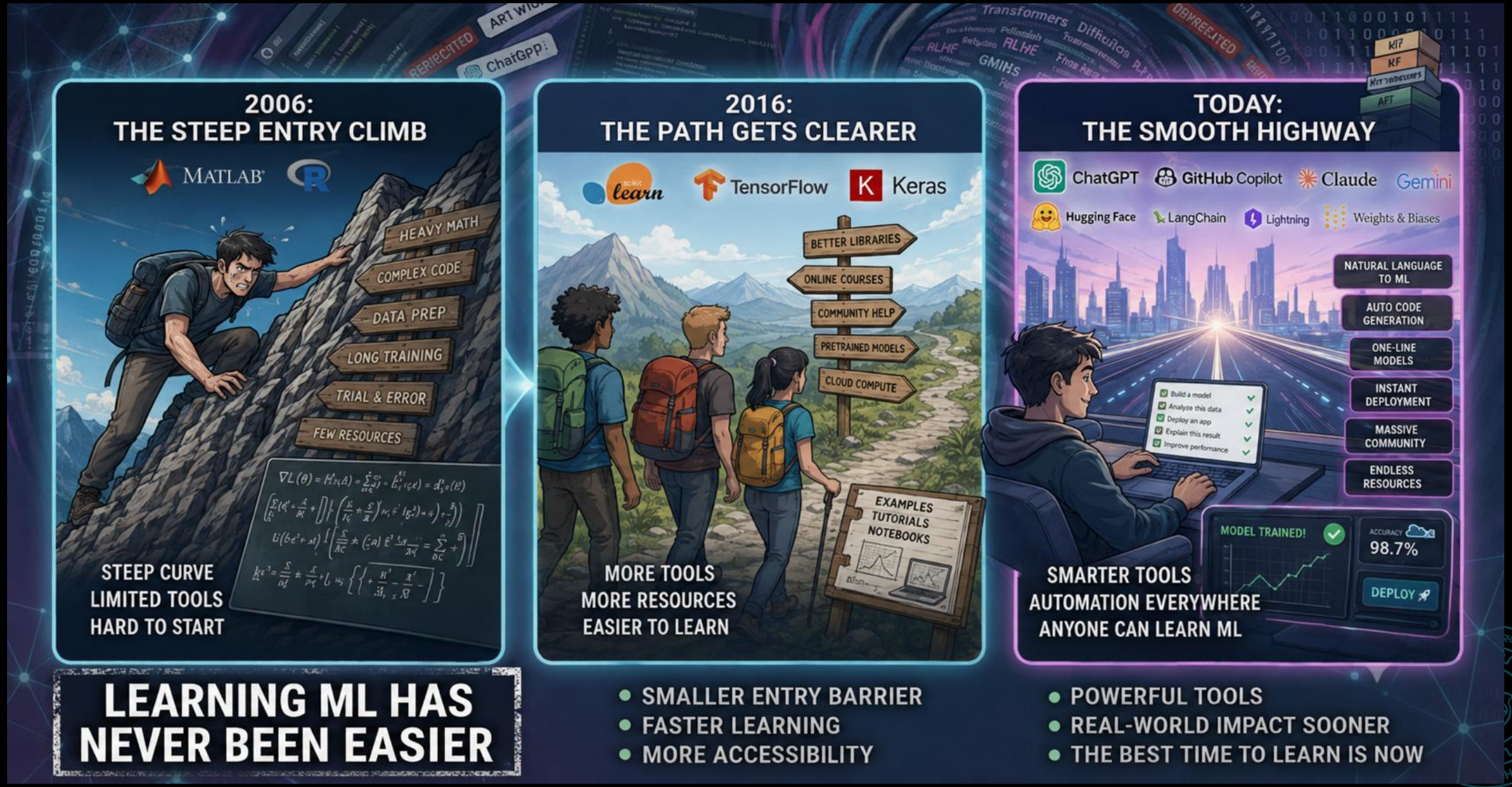
Wait! It's an AI Course or ML Course?

- Doesn't matter what you want to call it. 😊
- AI is a broader discipline
- ML and Deep Learning (DL) are subfields within AI
  - Both ML and DL involve learning from data
- Much of modern AI is largely powered by ML/DL techniques
- Thus understanding ML/DL is key to understanding modern AI
- This course is about ML (and some DL)

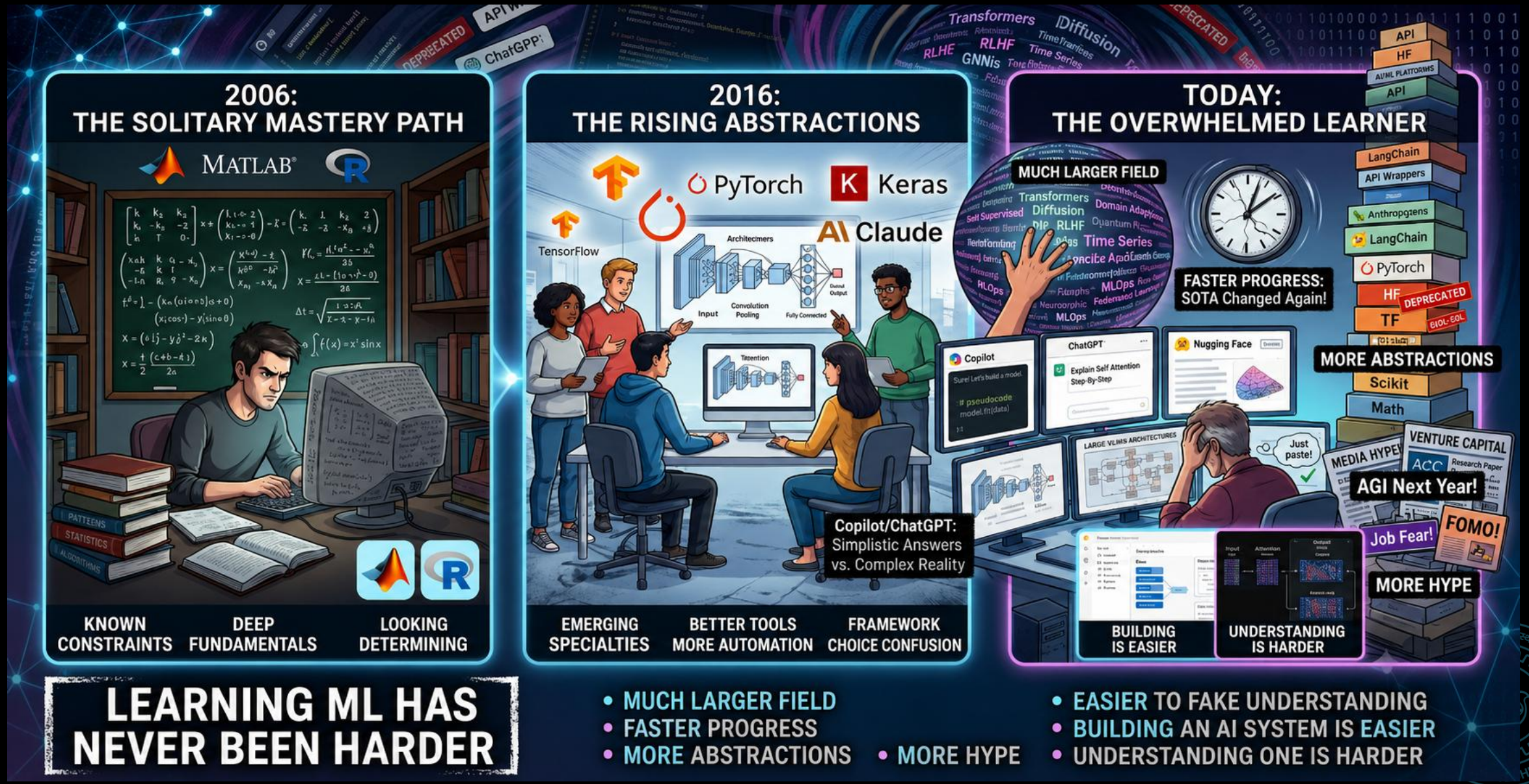
  ...with a taste of ML-driven AI advances.



In Today's Times, is Learning ML Easier or Harder?



In Today's Times, is Learning ML Easier or Harder?



In Today's Times, is Learning ML Easier or Harder?

Easier

- Accessible computing
  - GPUs, cloud, Colab
- Mature ML tools
  - PyTorch, Hugging Face
- Abundant learning resources
- Tutorials, repos, courses
- AI assistants
  - ChatGPT, Copilot

Harder

- Fast-moving field
  - Explosive growth
- More information, more hype
- Easier to copy or vibe-code
  - .. illusion of understanding
- Harder to ..
  - Build intuition
  - Know what works
  - Know when/why models fail

Getting into ML/AI ? Two Learning Paths..

A Tale of Two Students

Both paths have value. But deeper understanding creates greater impact.

Student A

I can use AI tools



- Write code
- Call APIs
- Fine-tune models
- Prompt models

VS.

Student B

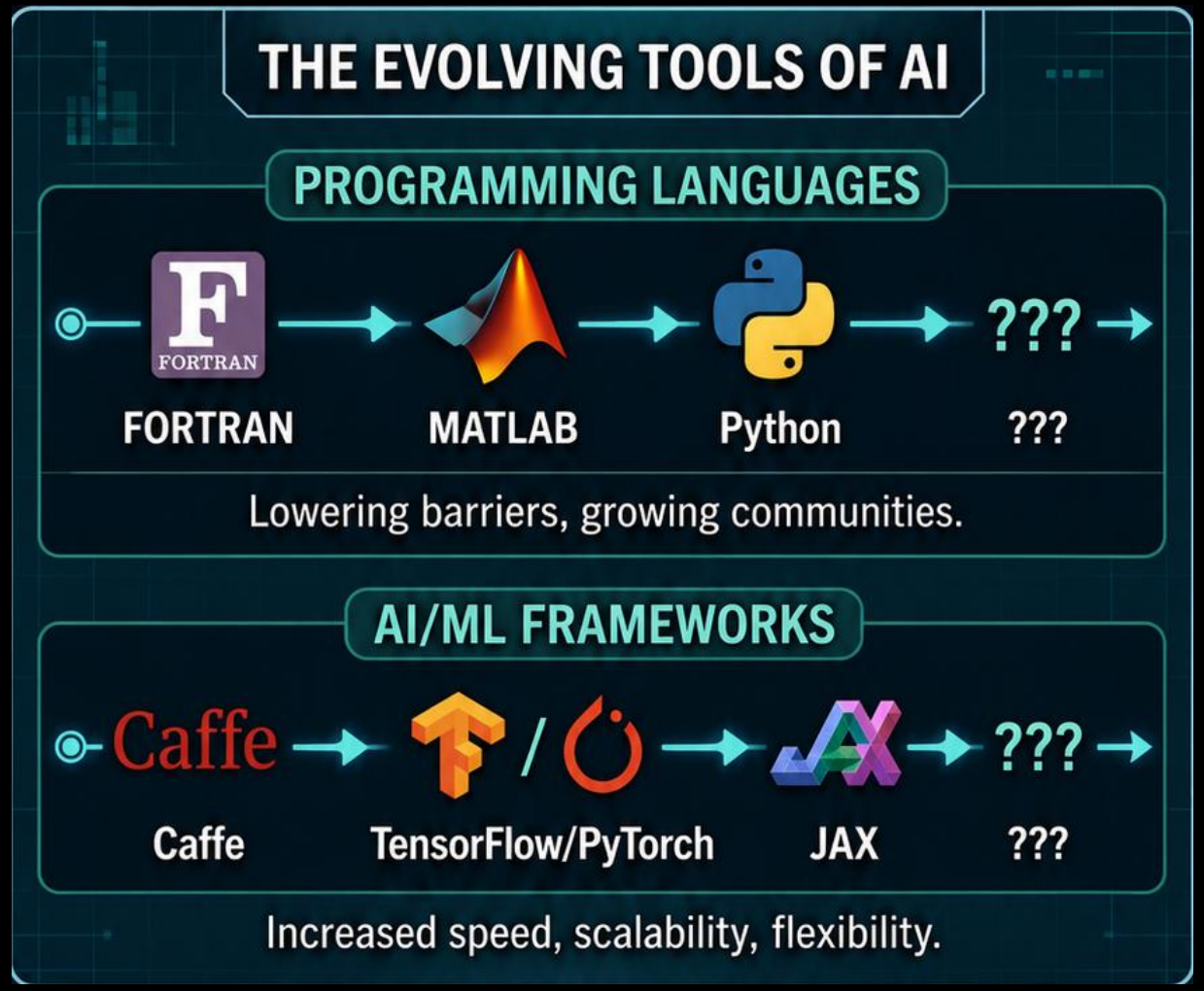
I understands the core principles

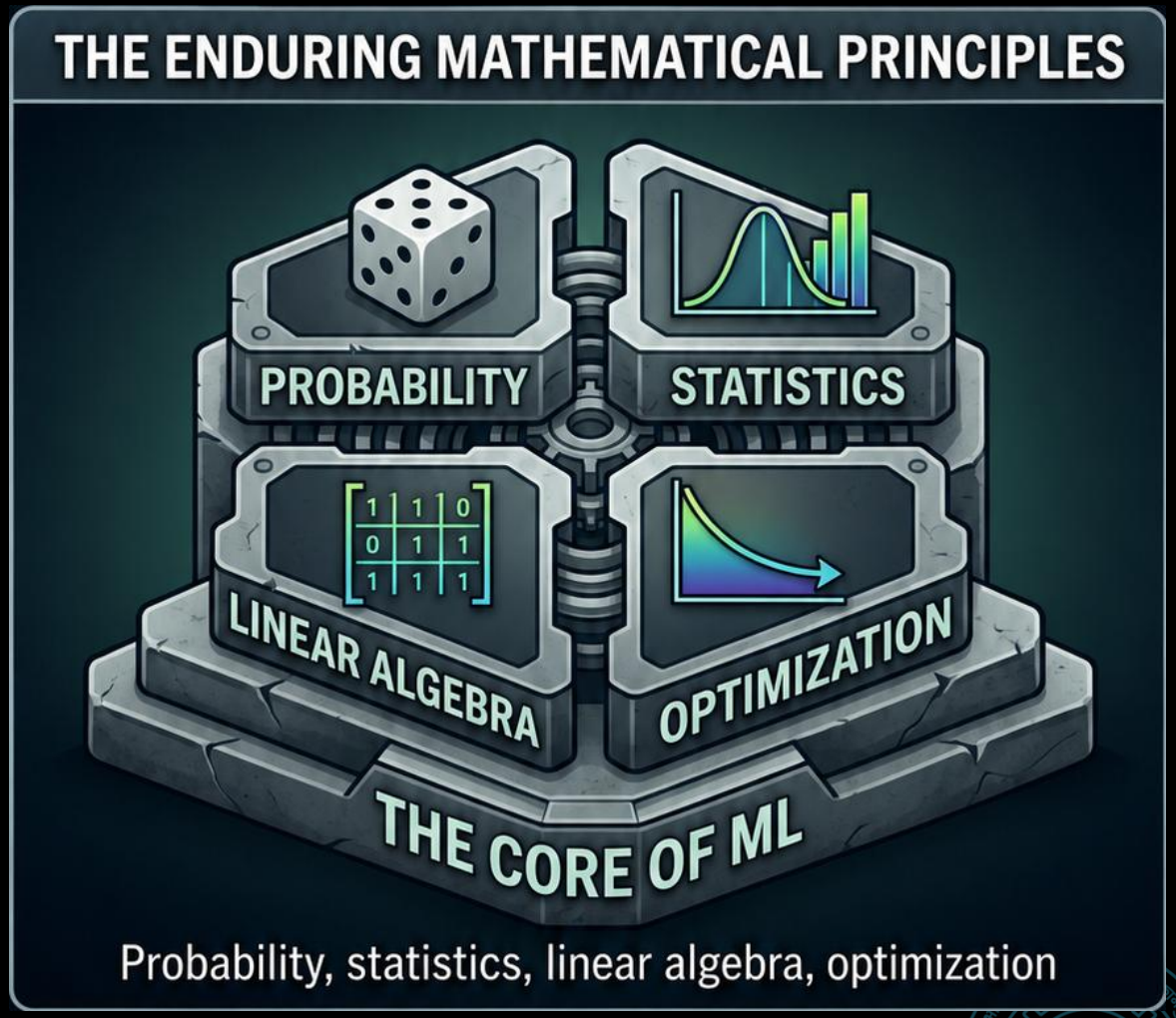


- Why it works
- Why it fails
- How to improve it
- When not to trust it

Principles outlive Tools

Remember: Principles Outlive Tools





Today's AI can even explain principles, but *you* have to *understand* them

What is Machine Learning?

Remember: ML cares about predictions on new, previously unseen inputs, not training inputs



Machine learning is the science of building algorithms that learn patterns from data and use what they learn to make predictions or decisions.

DATA



Examples from the real world
(images, text, signals, tables, ...)

+

ALGORITHM



A learning procedure that finds patterns in data

=

MODEL

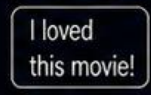
$f(x)$

A learned function that maps inputs to outputs

→

PREDICTIONS / DECISIONS


→ Cat



→ Positive


→ 42.7

⋮

Used on new, unseen data to make predictions or decisions

KEY IDEAS



Learn from data
Improve with more data and experience.



Find patterns
Discover relationships in data automatically.



Generalize
Work on new, unseen examples.



Make predictions
Provide useful outputs or decisions.



Improve over time
Get better as more data becomes available.

Highly desirable for an ML system

When is Machine Learning Necessary?

Use ML when we can't enumerate all possible rules to handle the complexity of real-world data.

Traditional Programming Struggles

We try to write explicit rules for every case.
Rules become too many, too complex, or impossible.

Machine Learning Helps

Learn patterns from data automatically.
Generalize to new, unseen situations.

Real-World Scenarios

Image Recognition



Example: Identify objects in photos (cat, car, plane)

Why rules don't scale:
Too many variations in lighting, pose, background, shape, occlusion, etc.

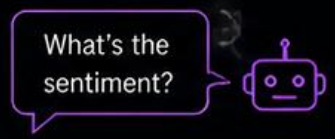
Speech Recognition



Example: Convert speech to text

Why rules don't scale:
Accents, speed, noise, overlapping speech, new words, context.

Natural Language Understanding



Example: Understand sentiment or intent

Why rules don't scale:
Language is creative, ambiguous, sarcastic, and ever-changing.

Recommendation Systems



Example: Recommend products or videos

Why rules don't scale:
User preferences are unique, evolve over time, and depend on context.

Fraud Detection



Example: Detect fraudulent transactions

Why rules don't scale:
Fraud patterns change constantly and are rare and subtle.

Medical Diagnosis

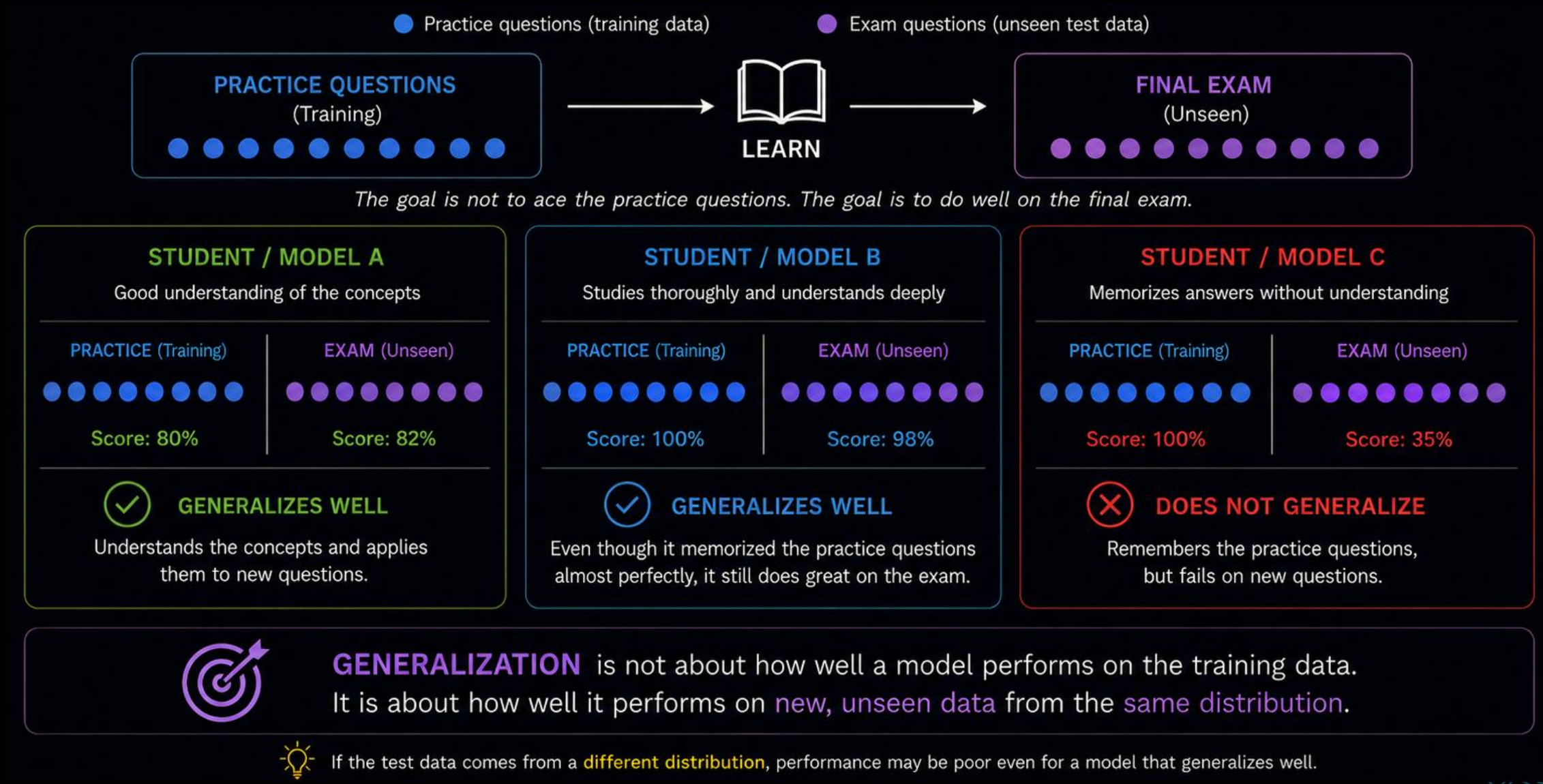


Example: Detect disease from medical scans

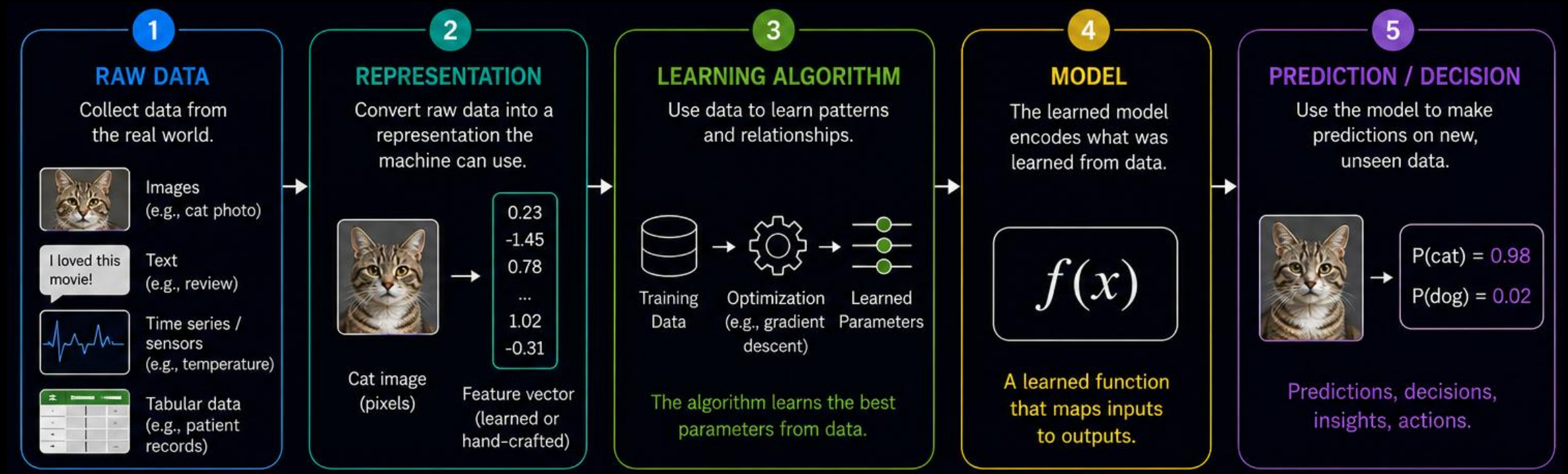
Why rules don't scale:
Countless appearance variations; new cases are hard to anticipate.

Rule-based systems work well for simple, well-defined problems. | Machine learning is essential for complex, uncertain, and dynamic real-world problems.

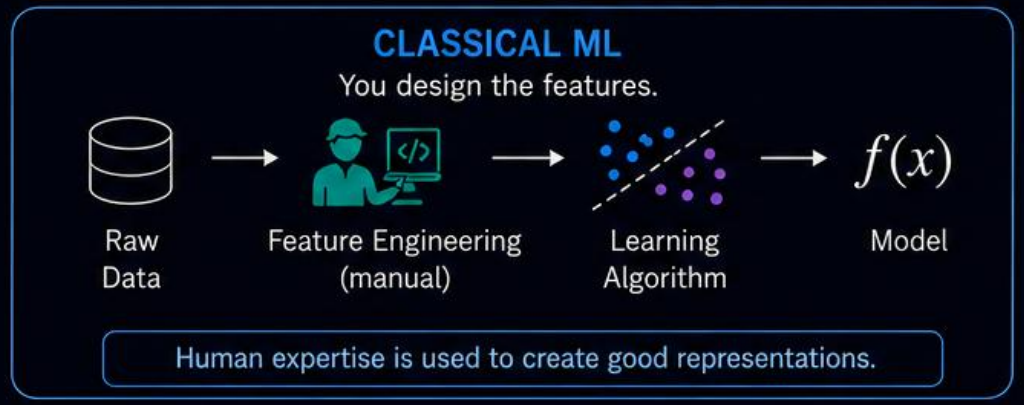
ML and The Principle of Generalization

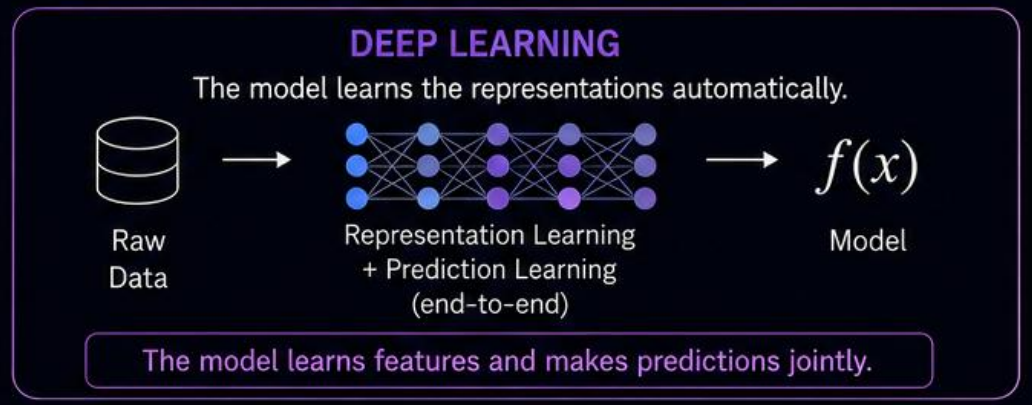


The Machine Learning Workflow



TWO BROAD APPROACHES TO REPRESENTATION AND LEARNING





Data as Geometric Primitives

ML algorithms represent and treat data as points in a high-dim space



Real-world object
(e.g., an image)

Machine
(feature extraction / representation)

Point in a high-dimensional space

In this space, nearby points are usually similar.


Wolf

Close


Dog

Moderately far


Cat

Far

Moderately far


Tiger


Car

Patterns, clusters and structure live in this geometry.

Once data is represented as points, machine learning discovers useful patterns and functions in this geometry.

Many core ideas in machine learning rely on this geometric view.

Nearest Neighbors



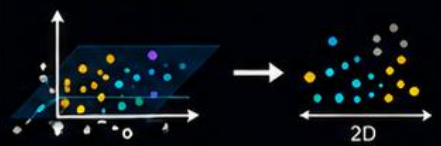
Use nearby points to make predictions.

Clustering



Group similar points together.

Dimensionality Reduction (e.g., PCA)



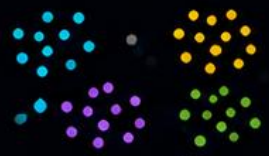
Find a lower-dimensional view that preserves structure.

Kernel Methods



Measure similarity via implicit geometry.

Embeddings



Learn representations where meaning is captured by geometry.

Transformers



Attention computes similarity between all pairs of points.

ML: Some Basic Problem Types

These are the most common ones. Several other ML problem types exist. Will study those later



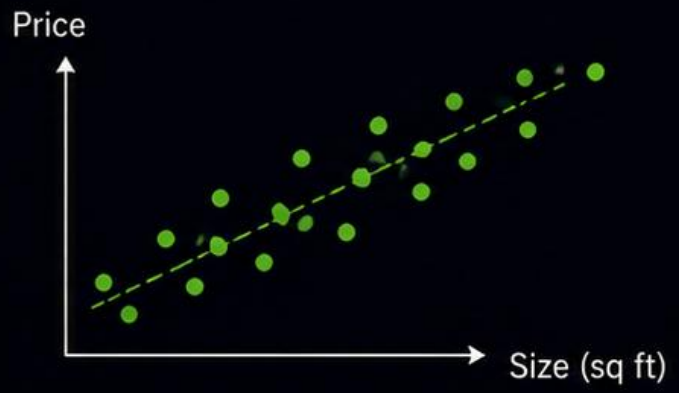
Machine learning algorithms learn from data to find patterns and make sense of the world.

SUPERVISED LEARNING

Learn from labeled examples (we know the answers)

1. REGRESSION

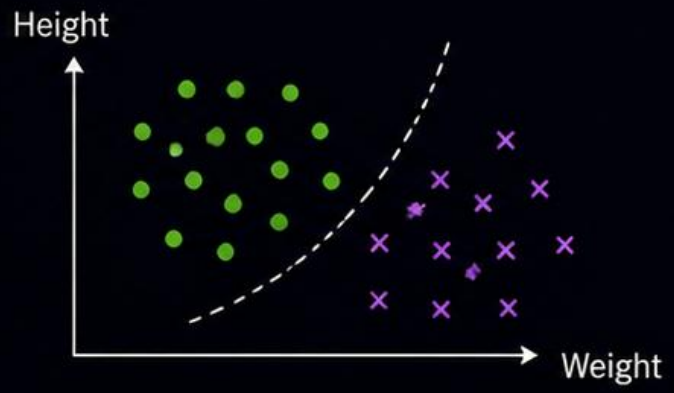
Predict a continuous value



Example: Predict the price of a house given its size.

2. CLASSIFICATION

Predict a class or category



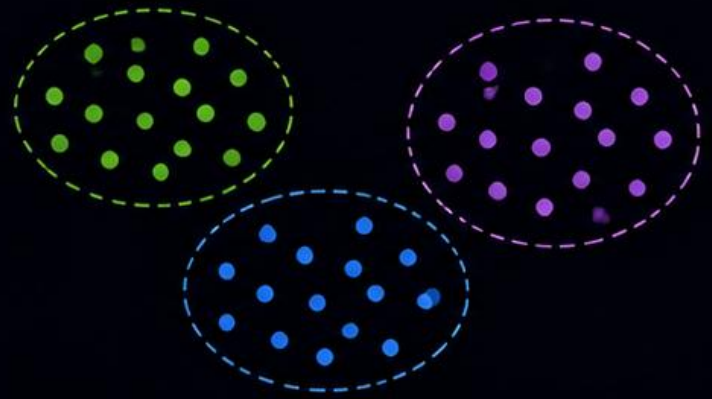
Example: Is this animal a cat or a dog?

UNSUPERVISED LEARNING

Find structure in unlabeled data (no answers given)

3. CLUSTERING

Discover groups or clusters



Example: Group customers with similar buying behavior.

Supervised learning uses known answers to make predictions. | Unsupervised learning finds hidden patterns on its own.

ML is Essentially Function Approximation

INPUT x (e.g., an image) → **TRAINING DATA** (known examples) (x_1, y_1) (x_2, y_2) ... (x_n, y_n) → **LEARN THE FUNCTION f** $y = f(x)$ A function f that maps an input x to an output y. → **PREDICT FOR NEW INPUT** (unseen example) x_{new} → **OUTPUT y** (e.g., "cat") "cat"

We have pairs (x, y) (labels available)

Use f to predict y (generalization)

SAME IDEA, MANY PROBLEMS

Note: RL isn't really supervised learning; but more on this later

SUPERVISED LEARNING (Labels available)

1 REGRESSION
Predict a continuous value.
₹ 85,00,000
Price (a number)

2 CLASSIFICATION
Predict a discrete label.
Dog
Label (cat / dog)

3 LANGUAGE MODELING
Predict the next token/word.
The quick brown fox → jumps
Previous words (context) | Next word (a token)

4 REINFORCEMENT LEARNING
Predict the best action.
State (s) | Action (a) → right

Goal: learn a function f that maps inputs x to outputs y.
(Learn $f : \mathcal{X} \to \mathcal{Y}$)

UNSUPERVISED LEARNING (No labels)

1 REPRESENTATION LEARNING
Learn a function that maps x to useful representations z.
z (embedding)
Learn $f : \mathcal{X} \to \mathcal{Z}$

2 CLUSTERING
Learn a function that assigns each x to a cluster.
Cluster ID
Learn $f : \mathcal{X} \to \{1, \ldots, K\}$

3 DIMENSIONALITY REDUCTION
Learn a function that maps x to a lower-dimensional representation z.
High-dim x (3D) → Low-dim z (2D)
Learn $f : \mathcal{X} \to \mathbb{R}^k$ $(k \ll d)$

Goal: learn a function f that reveals structure in the data.
(Learn $f : \mathcal{X} \to \mathcal{Z}$ or $f : \mathcal{X} \to \mathcal{Y}$) where labels are not given)

Different ML algorithms differ primarily in the family of functions they can learn.

Linear Models | Decision Trees | SVMs | Kernel Methods | Neural Networks | Transformers (Attention) | ...

Different families of functions

The ML Approach for Some Broad Problems

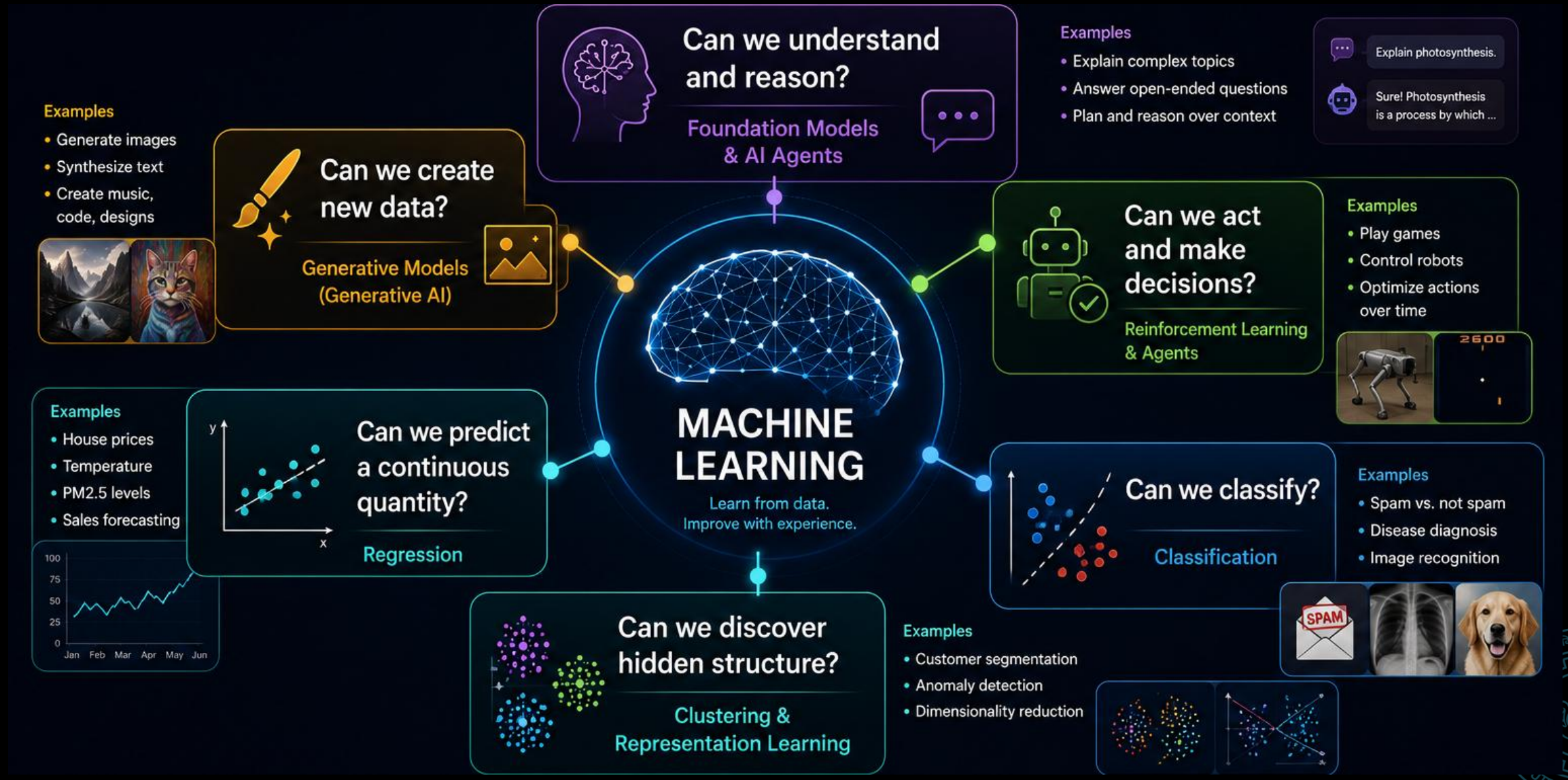


Illustration: Some Problems and ML Solutions

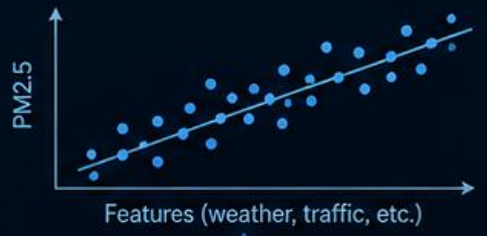
The Problem

What will be the PM2.5 tomorrow in each location?



Same real-world problem.
Different questions.
Different ML models/solutions.

1 Can we predict a continuous quantity?



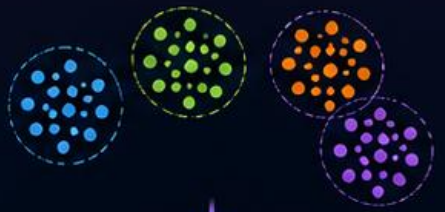
Regression
e.g., Linear Regression, SVR, Gradient Boosting

2 Will AQI be hazardous?



Classification
e.g., Logistic Regression, SVM, Trees, Neural Nets

3 Which locations behave similary?



Clustering
e.g., K-means, GMM, Spectral Clustering

4 What factors drive pollution?



Explainability
e.g., Decision Trees, SHAP, Feature Importance

5 Can we forecast next week?



Time-Series Forecasting
e.g., ARIMA, LSTM, TCN, Transformers

6 What actions reduce pollution best?



RL / Agents
e.g., RL, Causal ML, Optimization, Multi-agent RL

Illustration: Some Problems and ML Solutions

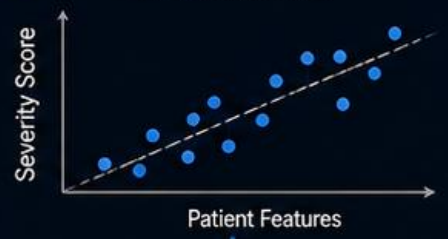
The Problem

Given patient data and scans, how can we understand and treat the disease?



Same patient.
Different questions.
Different ML models/solutions.

1 How severe is the disease?



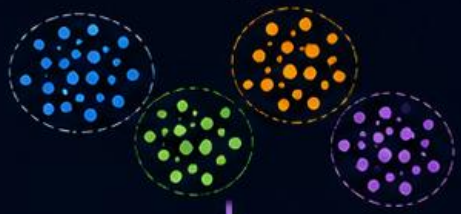
Regression
e.g., Linear Regression, SVR, Neural Networks

2 Is it benign or malignant?



Classification
e.g., Logistic Regression, SVM, Trees, Neural Nets

3 Are there unknown subtypes?



Clustering
e.g., K-means, GMM, Hierarchical Clustering

4 Which factors are most important?



Explainability
e.g., Decision Trees, SHAP, Feature Importance

5 Can we generate a radiology report?



Generative Models
e.g., LLMs, Seq2Seq, Diffusion Models

6 What treatment should we recommend?

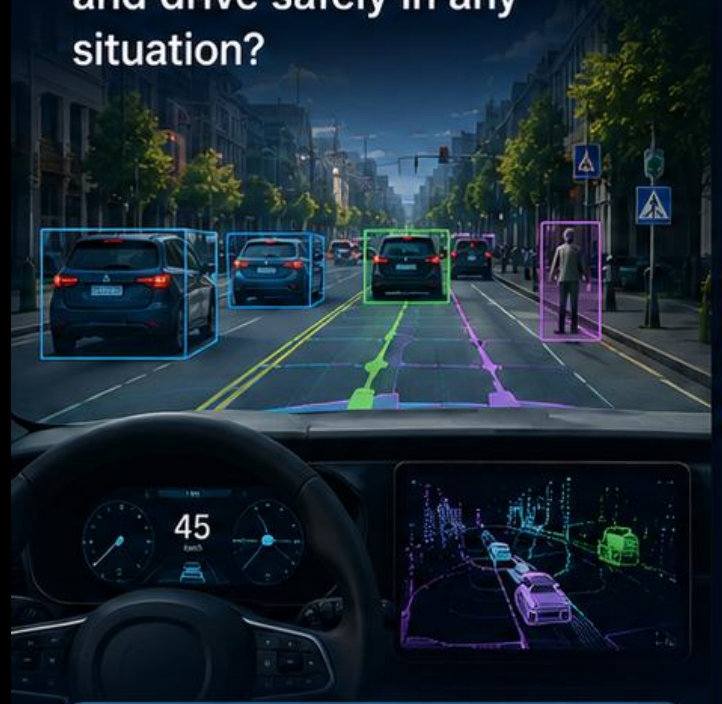


RL / Agents
e.g., RL, Bandits, Decision Policies

Illustration: Some Problems and ML Solutions

The Problem

How do we perceive, predict, and drive safely in any situation?



Same driving scenario.
Different questions.
Different ML models/solutions.

1 What is the lane position?



Regression
e.g., Linear Regression, Kalman Filter, NN

2 Is this a pedestrian or a vehicle?



Classification
e.g., SVM, Trees, CNNs, Transformers

3 How can we understand the scene?



Representation Learning
e.g., Autoencoders, Contrastive Learning, GNNs

4 Where will objects move next?



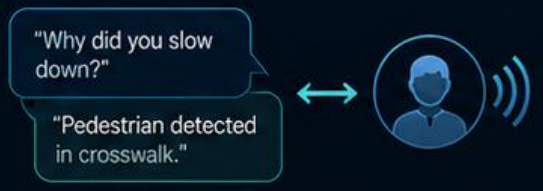
Forecasting
e.g., LSTM, GNNs, Transformers, Kalman Filter

5 What is the best action to take?



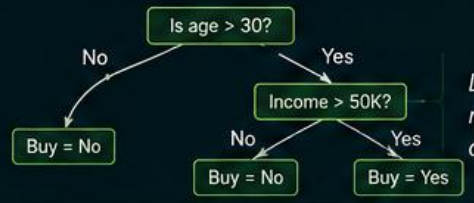
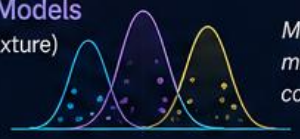
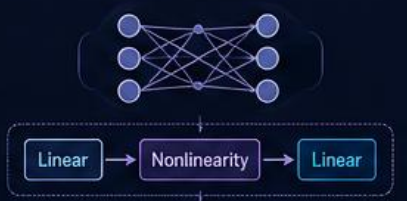
RL / Decision Making
e.g., RL, Imitation Learning, Optimal Control

6 Can we explain and interact with the car?



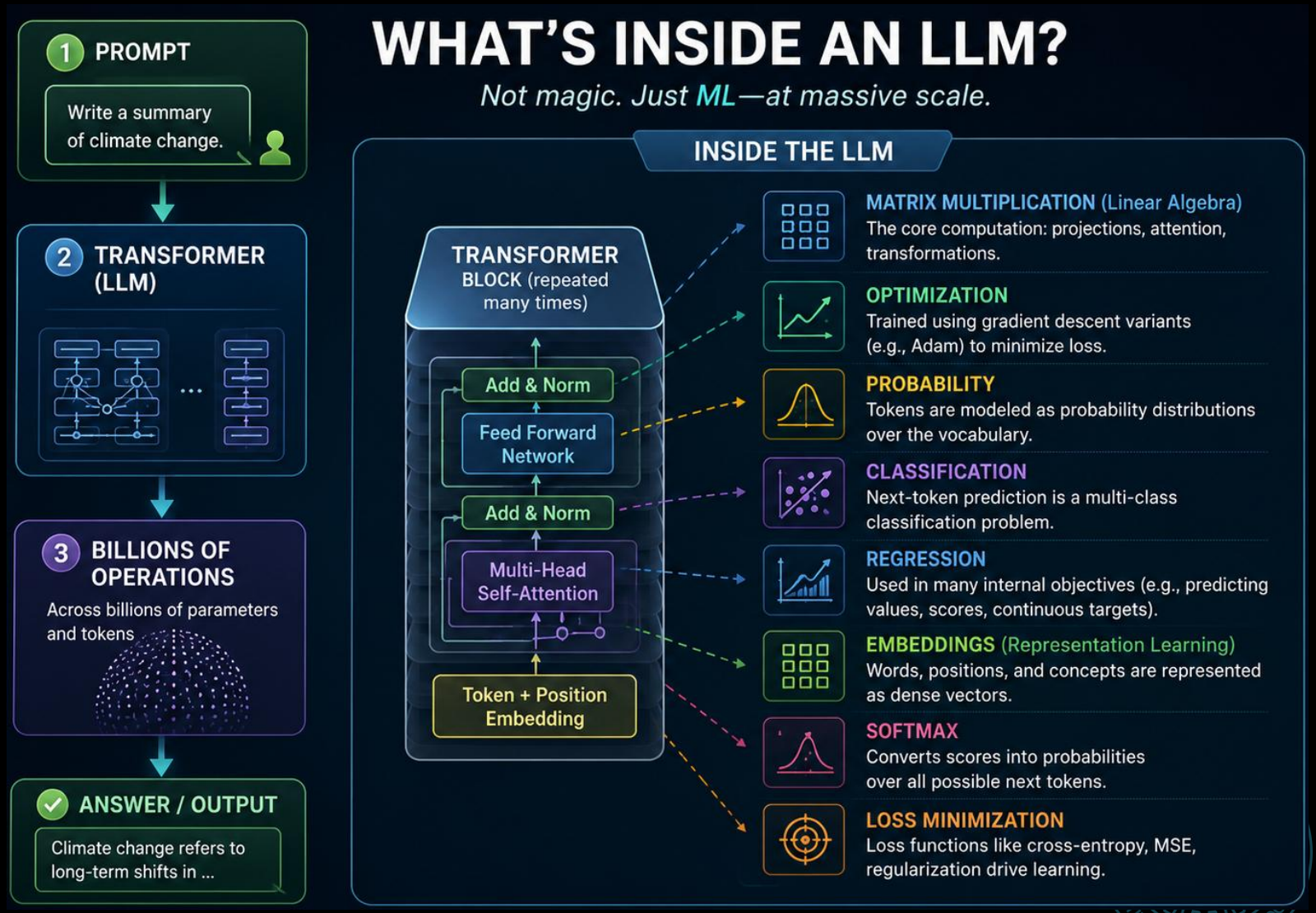
LLMs / Agents
e.g., VLMs, LLMs, Planning Agents, Tool Use

Classical ML Topics and Their Relevance Today

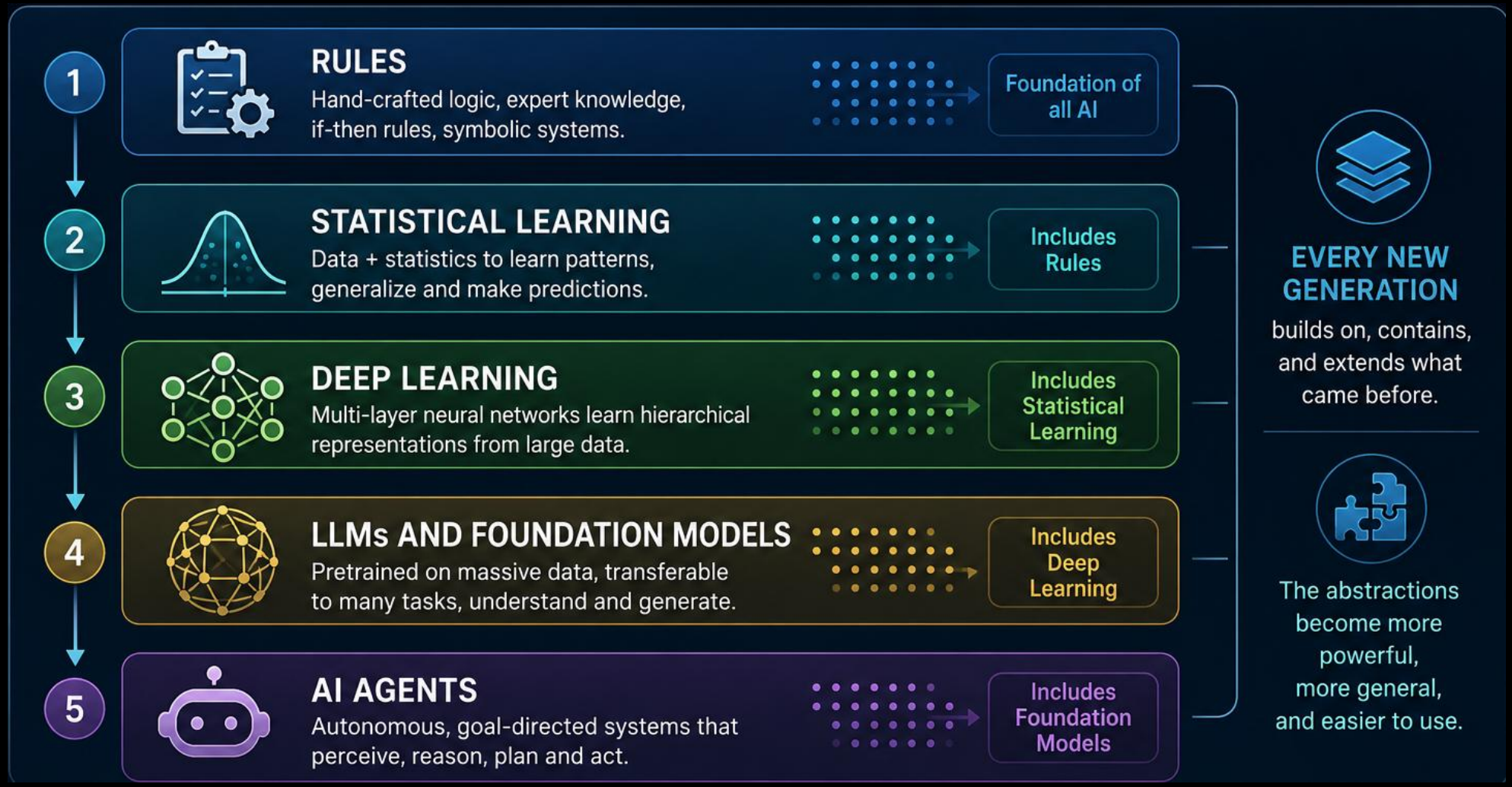
WHAT WE LEARN	WHY IT MATTERS	TODAY: WHERE IT POWERS MODERN AI
1 Learning from Similarity (Nearest Neighbors)  *Predict from similar examples*	Learn from similar examples Basis of retrieval and memory-based AI	**TODAY** Retrieval-Augmented Generation (RAG) Vector databases (FAISS, Pinecone, Milvus) Search — Semantic search and recommendation 
2 Learning Decision Rules (Decision Trees)  *Learn simple rules to make decisions*	Interpretable decision making Sequential rule learning	**TODAY** Explainable AI and rule extraction Boosted trees (XGBoost, LightGBM) Model interpretation & feature importance 
3 Learning Functions (Linear Models)  *Fit a function to predict continuous values* + **Basic Mixture Models** (e.g., Gaussian Mixture)  *Model data as a mixture of simple components*	Predict by learning mathematical functions Simplest useful predictive models Capture multi-modal structure and heterogeneity	**TODAY** Linear probes for understanding LLMs Calibration and uncertainty estimation Interpretable models and forecasting **Deep neural networks** Built from linear + nonlinearity blocks **Mixture of Experts (MoE) & mixture-based deep learning** Specialize experts; combine intelligently  MoE 

Why Study Basic ML and Not Just Study LLMs?

- LLMs are built on basic and fundamental ML ideas
- Many real-world problems are still best solved by the basic ML ideas
- ML fundamentals help us build and understand
- Understanding help develop intuition too

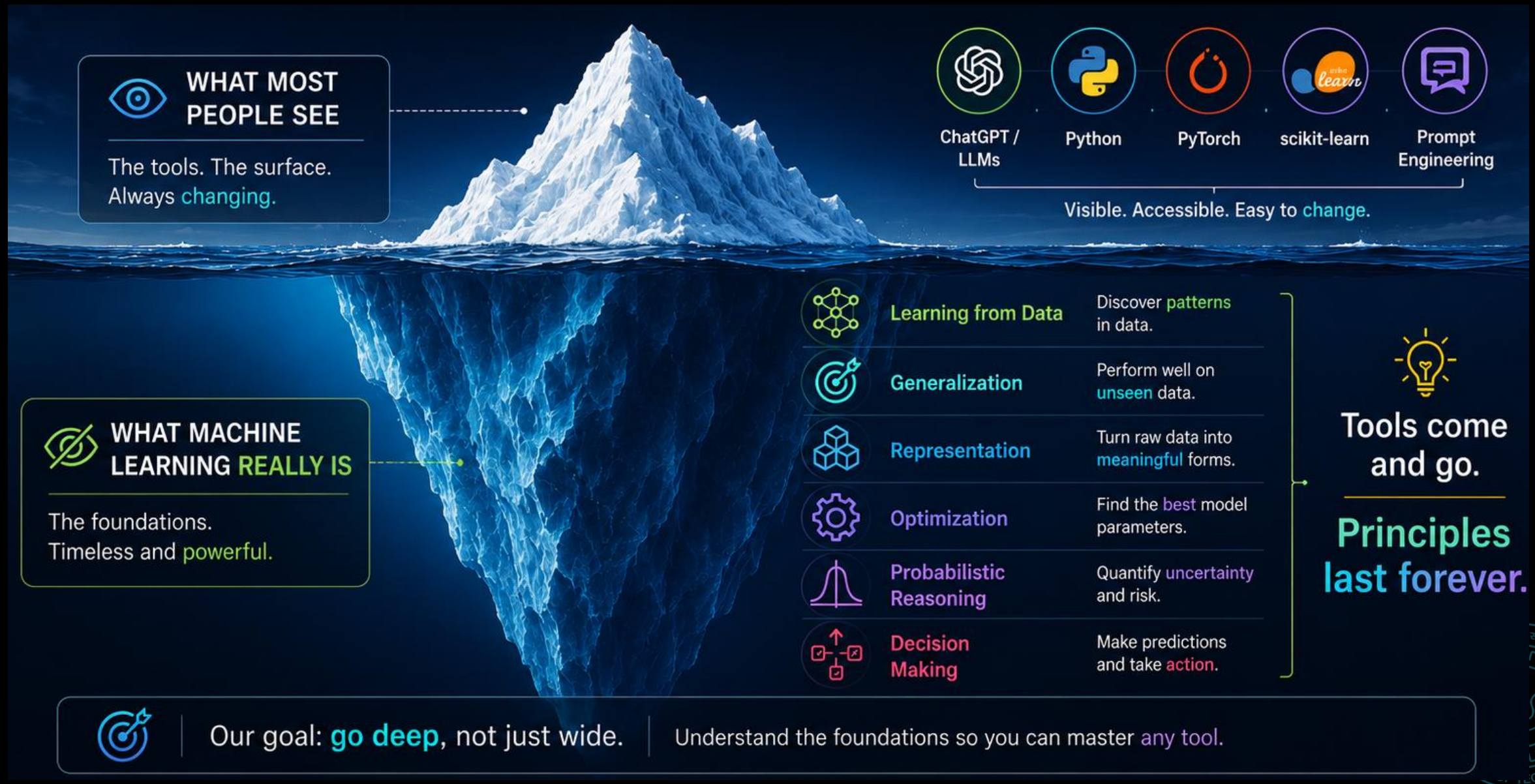


Evolution of ML/AI Over The Years..

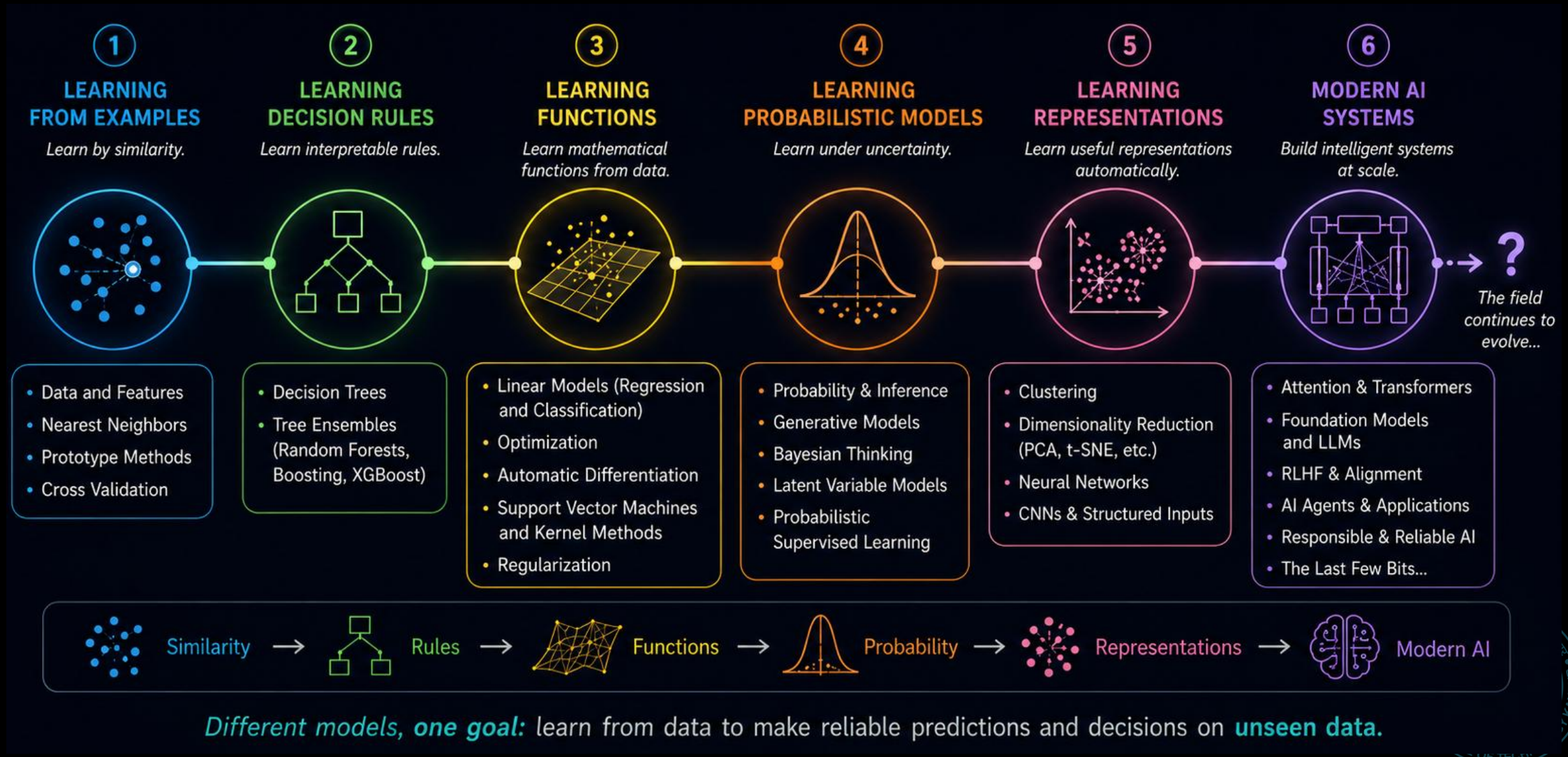


ML/AI evolution has been additive, not destructive. Each step raised the level of abstraction but relied upon the same core foundations

ML is (Still) More Than Just Software Tools/APIs



Our Upcoming ML Journey in CS771



Next Class

- Data, Features, Data Geometry, and Operations on Data
- Learning by computing similarities

